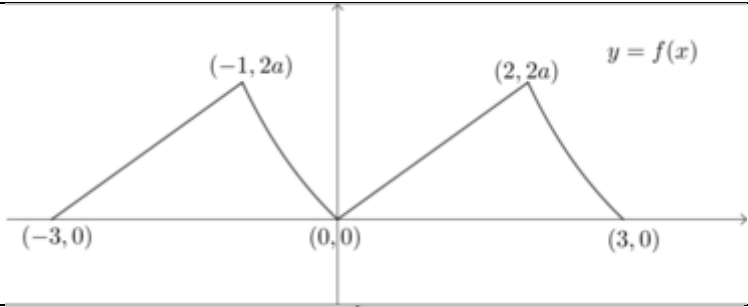
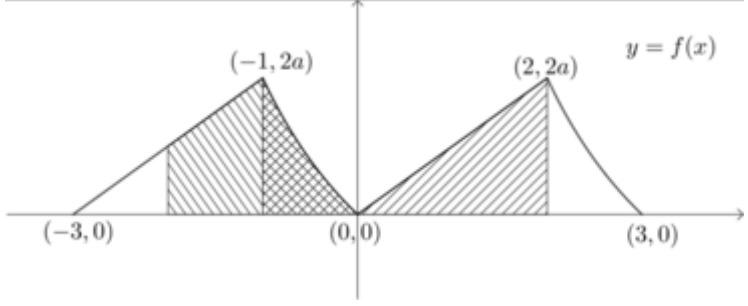
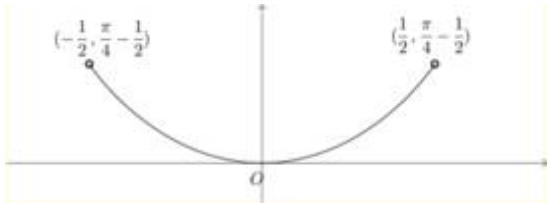


2019 NJC Preliminary Examination H2 Mathematics Paper 2 Marker's Comments

1	Suggested Solution	Marker's Comments
(i)	$f\left(\frac{44}{3}\right) = f\left(\frac{8}{3} + 3 \times 4\right) = f\left(\frac{8}{3}\right) = \frac{12a}{\frac{8}{3}} - 4a = \frac{a}{2}$	Note the clear and efficient way of showing that the graph repeats itself after 3 units i.e. $f\left(\frac{44}{3}\right) = f\left(\frac{8}{3} + 3 \times 4\right)$.
(ii)		Common mistakes: - hollow point for (3,0) - since graph has a period of 3 units, the maximum point should have x-coordinate -1 and 2. Please draw all graphs with a dark pencil.
(ii)	 $\begin{aligned} \int_{-2}^2 f(x) dx &= \frac{1}{2}(2)(2a) + \frac{1}{2}(1)(a + 2a) + \int_{-1}^0 f(x) dx \\ &= 2a + \frac{3}{2}a + \int_2^3 f(x) dx \\ &= \frac{7}{2}a + \int_2^3 \frac{12a}{x} - 4a dx \\ &= \frac{7}{2}a + [12a \ln x - 4ax]_2^3 \\ &= \frac{7}{2}a + 12a \ln 3 - 12a - 12a \ln 2 + 8a \\ &= 12a \ln\left(\frac{3}{2}\right) - \frac{1}{2}a \end{aligned}$	Many students did not manage to come out with this expression $\int_2^3 \frac{12a}{x} - 4a dx$. Due to the graph having a period of 3 units, the curve between $x = -1$ and $x = 0$ will appear to be the same as in the interval $x = 2$ to $x = 3$. But this does not mean that $\int_2^3 \frac{12a}{x} - 4a dx = \int_{-1}^0 \frac{12a}{x} - 4a dx$. Be careful and notice that there are three partitions also.

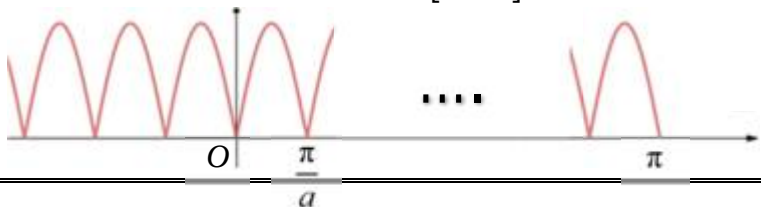
2	Suggested Solution	Marker's Comments
(i)	$\frac{d^2y}{dx^2} = \frac{2}{\sqrt{1-4x^2}}$ $\frac{dy}{dx} = \int \frac{2}{\sqrt{1-4x^2}} dx = \int \frac{2}{\sqrt{1-(2x)^2}} dx = \sin^{-1}(2x) + C$ $y = \int \sin^{-1}(2x) + C dx$ $= \int 1 \cdot \sin^{-1}(2x) dx + Cx + D$ $= x \sin^{-1}(2x) - \int x \left(\frac{2}{\sqrt{1-4x^2}} \right) dx + Cx + D$ $= x \sin^{-1}(2x) + \frac{1}{4} \int (-8x)(1-4x^2)^{-\frac{1}{2}} dx + Cx + D$ $= x \sin^{-1}(2x) + \frac{1}{4} \frac{(1-4x^2)^{\frac{1}{2}}}{-\frac{1}{2}} + Cx + D$ $= x \sin^{-1}(2x) + \frac{1}{2} \sqrt{1-4x^2} + Cx + D$	Most students knew that they had to refer to MF26 to integrate $\frac{2}{\sqrt{1-(2x)^2}}$. However, many mistakes were made. Common mistakes: - When using the formula for integrating $\frac{1}{\sqrt{1-x^2}}$ to integrate $\frac{1}{\sqrt{1-(2x)^2}}$, students forgot to divide by 2. - When taking out factor of 4 from $\sqrt{1-4x^2}$, students forgot to square root the 4. - When differentiating $\sin^{-1}(2x)$, many students gave $\frac{1}{\sqrt{1-(2x)^2}}$ instead of $\frac{2}{\sqrt{1-(2x)^2}}$. There were many other minor mistakes in algebraic manipulation.
(b) (i)	When $x=0$, $y=0$ and $\frac{dy}{dx}=0$. $0 = \sin^{-1}(0) + C \Rightarrow C=0$. $0 = 0 \sin^{-1}(0) + \frac{1}{2} \sqrt{1-0} + C(0) + D \Rightarrow D = -\frac{1}{2}$ $f(x) = x \sin^{-1}(2x) + \frac{1}{2} \sqrt{1-4x^2} - \frac{1}{2}$	There were quite a lot of students who did not seem to understand the term “particular solution curve”. They tried to “guess” an equation of a curve that has a minimum point at the origin, often giving the equation $y = kx^2$.
(b) (ii)	$\frac{1}{2} \sin^{-1} 1 + \frac{1}{2} \sqrt{1-4\left(\frac{1}{2}\right)^2} - \frac{1}{2} = \frac{\pi}{4} - \frac{1}{2}$ 	Most students did not label the end points, or did not bother to indicate the domain.

3	Suggested Solution	Marker's Comments
(a) (i)	Let $(x + yi)^2 = 8 - 6i$, $x, y \in \mathbb{R}$. $x^2 + 2xyi - y^2 = 8 - 6i$ $\begin{cases} x^2 - y^2 = 8 & \dots(1) \\ 2xy = -6 & \Rightarrow y = -\frac{3}{x} \dots(2) \end{cases}$ Substitute (2) into (1), $x^2 - \frac{9}{x^2} = 8$ $x^4 - 8x^2 - 9 = 0$ $(x^2 - 9)(x^2 + 1) = 0$ $x^2 = 9 \text{ or } x^2 = -1 \text{ (rejected)}$ $x = 3 \text{ or } x = -3$ Correspondingly, $y = -1$ or $y = 1$. $x = 3, y = -1 \text{ or } x = -3, y = 1$	Question 3 is poorly performed in general. One common issue observed is that students missed or ignored the instruction "Do not use a calculator ..." at the start of the question. Detailed steps are expected but are often missing. Most of the students were able to set simultaneous equations by comparing coefficient, but only a small proportion could produce a correct full solution. Common Mistakes: 1. Expand $(x + yi)$ wrongly. 2. Make x or y in terms of the other wrongly from $2xy = -6$ 3. Use GC to solve $x^4 - 8x^2 - 9 = 0$ or write down the roots without working. 4. Reject a non-real root (probably after comparing with output from a GC). Comparing non-real numbers is incorrect and a non-real number has two square roots.
(a) (ii)	$w^2 + 2iw - 9 + 6i = 0$ $w^2 + 2iw = 9 - 6i$ $w^2 + 2iw + i^2 = 9 - 6i + i^2$ $(w + i)^2 = 8 - 6i$ From (i), $w + i = 3 - i$ or $w + i = -3 + i$ $w = 3 - 2i \text{ or } w = -3$	This part is very poorly attempted. Most of the students did not show the correct approach (applying quadratic formula or completing the square) to solve the quadratic equation in w. Comparing real and imaginary parts is a tedious method and many students wasted a lot of time on it but cannot get the answer. An alarming number of students cannot recall the quadratic formula correctly. Be exam smart: 1. It is not likely that two parts from the same question are testing the same concept/skill (in this case, comparing real and imaginary parts). 2. It is likely that part (i) and part (ii) are related in some ways.
(b) (i)	Substitute $z = -z_1$ into $z^4 - pz^2 + q$.	This part is poorly attempted. Many students did not show the proper presentation to verify a root.

	$(-z_1)^4 - p(-z_1)^2 + q$ $= z_1^4 - pz_1^2 + q = 0 \text{ (since } z_1 \text{ is a root)}$ Or $z^2 = (-z_1)^2, z^4 = (-z_1)^4$ Therefore $-z_1$ is also a root.	By merely writing $(-z_1)^4 = z_1^4, (-z_1)^2 = z_1^2$, or $(-z_1)^4 - p(-z_1)^2 + q = z_1^4 - pz_1^2 + q$, one failed to show the expression is equal to 0. Many students could not distinguish between $-z_1$ and z_1^*, and applied the Conjugate Root Theorem though the question does not give the condition that the coefficients are real.
(b) (ii)	$z = 3e^{i\left(\frac{\pi}{6}\right)}$ $z^2 = \left[3e^{i\left(\frac{\pi}{6}\right)}\right]^2 = 9e^{i\left(\frac{\pi}{3}\right)} = 9\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 9\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ $z^4 = \left[3e^{i\left(\frac{\pi}{6}\right)}\right]^4 = 81e^{i\left(\frac{2\pi}{3}\right)} = 81\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 81\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$ Substitute these values into $z^4 - 9z^2 + 81$ $81\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) - 9\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + 81$ $= -\frac{81}{2} + \frac{81\sqrt{3}}{2}i - \frac{9}{2} - \frac{9\sqrt{3}}{2}i + 81$ $= 0$ Thus $3e^{i\left(\frac{\pi}{6}\right)}$ is a root. By (b)(i) and conjugate root theorem, $3e^{i\left(-\frac{\pi}{6}\right)}, -3e^{i\left(\frac{\pi}{6}\right)}, -3e^{i\left(-\frac{\pi}{6}\right)}$ are also the roots of this equation. One quadratic factor is $\left[z - 3e^{i\left(\frac{\pi}{6}\right)}\right]\left[z - 3e^{i\left(-\frac{\pi}{6}\right)}\right] = z^2 - \left(3e^{i\left(\frac{\pi}{6}\right)} + 3e^{i\left(-\frac{\pi}{6}\right)}\right)z + 3e^{i\left(\frac{\pi}{6}\right)}3e^{i\left(-\frac{\pi}{6}\right)}$ $= z^2 - 2\left[3\cos\left(\frac{\pi}{6}\right)\right]z + 9$ $= z^2 - 3\sqrt{3}z + 9$ And the other quadratic factor is	This part is poorly attempted. Common Mistakes: 1. Convert z from its exponential form to Cartesian form, which should be discouraged as it is much easier to evaluate powers of a complex number in exponential form. 2. Use a GC (which is not allowed) or do not show sufficient working to convince the marker that a calculator is not used in the calculations. 3. Present a verification in a poor manner. To show LHS=RHS (0 in this case), one is not supposed to start with LHS=0 and eventually reach 0=0 after manipulations. Most of the students were unable to identify the other three roots (jointly from b(i) and Conjugate Root Theorem). Among those who managed to identify the roots, common mistakes are: 4. Fail to recall the Factor Theorem correctly. 5. Fail to pair up the linear factors to obtain quadratic factors with real coefficients as requested by question. 6. Fail to perform long-division correctly.

$ \begin{aligned} \left[z + 3e^{i\left(\frac{\pi}{6}\right)} \right] \left[z + 3e^{i\left(-\frac{\pi}{6}\right)} \right] &= z^2 + \left[3e^{i\left(\frac{\pi}{6}\right)} + 3e^{i\left(-\frac{\pi}{6}\right)} \right] z + 3e^{i\left(\frac{\pi}{6}\right)} 3e^{i\left(-\frac{\pi}{6}\right)} \\ &= z^2 + 2 \left[3 \cos \left(\frac{\pi}{3} \right) \right] z + 9 \\ &= z^2 + 3\sqrt{3}z + 9 \end{aligned} $	
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4	Suggested Solution	Marker's Comments
	$ \begin{aligned} \int_{-\pi}^{\pi} \sin kx \, dx &= \left(-\frac{\cos kx}{k} \right)_{-\pi}^{\pi} \\ &= \left(-\frac{\cos k\pi}{k} \right) - \left[-\frac{\cos(-k\pi)}{k} \right] \\ &= \frac{\cos(-k\pi)}{k} - \frac{\cos k\pi}{k} \\ &= 0 \quad (\because \cos A = \cos(-A)) \end{aligned} $ $ \begin{aligned} \int_{-\pi}^{\pi} \cos kx \, dx &= \left(\frac{\sin kx}{k} \right)_{-\pi}^{\pi} \\ &= \left(\frac{\sin k\pi}{k} \right) - \left[\frac{\sin(-k\pi)}{k} \right] \\ &= 0 - 0 \\ &= 0 \end{aligned} $	- Some students write $\int_{-\pi}^{\pi} \sin kx \, dx = (-k \cos kx)_{-\pi}^{\pi}$, which is incorrect - Many students didn't use the fact that $\cos(-k\pi) = \cos(k\pi)$ to conclude that $\cos(-k\pi) - \cos(k\pi) = 0$. Instead they seem to think that $\cos(-k\pi) = -1$, $\cos(k\pi) = 1$ which might not be true because it depends on the value of k. - Many students left $\int_{-\pi}^{\pi} \cos kx \, dx = \frac{2 \sin k\pi}{k}$ and didn't seem to realize that $\sin k\pi = 0$ for all integers k.

(i)	$\sin(mx)\sin(nx) = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x]$ $\int_{-\pi}^{\pi} \sin(mx)\sin(nx) dx$ $= \int_{-\pi}^{\pi} \frac{1}{2}[\cos(m-n)x - \cos(m+n)x] dx$ $= \frac{1}{2} \int_{-\pi}^{\pi} \cos(m-n)x dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos(m+n)x dx$ $= \frac{1}{2} \int_{-\pi}^{\pi} \cos(m-n)x dx - \frac{1}{2}(0)$ If $m \neq n$, $\int_{-\pi}^{\pi} \cos(m-n)x dx = 0$ using the first part result. If $m = n$, $\int_{-\pi}^{\pi} \cos(m-n)x dx = \int_{-\pi}^{\pi} 1 dx = \pi - (-\pi) = 2\pi$ Therefore, $\int_{-\pi}^{\pi} \sin(mx)\sin(nx) dx = \begin{cases} \pi & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$	• Not well-done. • Generally, students used factor formula correctly to arrive at $\int_{-\pi}^{\pi} \frac{1}{2}[\cos(m-n)x - \cos(m+n)x] dx$. However, most students didn't realize that the result in part (i) can be used to solve the subsequent parts. • A lot of students continued with integrating the above term and obtain $\frac{1}{2} \left[\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n} \right]$. This is then followed by the consideration of the different cases $m \neq n$ and $m = n$. Those that do this approach usually get stuck in the latter case. • Another large portion of students considered $m = n$ case first and deduced that the integral can be solved with a double angle formula, but many were careless with the factor of $\frac{1}{2m}$ (or $\frac{1}{2n}$) when integrating $\cos(2mx)$. • There are a few who used integration by parts, but they should realize that such approach should be unlikely given the allocation of marks.
(ii)	$\cos(mx + nx) = \cos(mx)\cos(nx) - \sin(mx)\sin(nx)$ $\int_{-\pi}^{\pi} \cos(mx + nx) dx$ $= \int_{-\pi}^{\pi} \cos(mx)\cos(nx) dx - \int_{-\pi}^{\pi} \sin(mx)\sin(nx) dx$ LHS = $\int_{-\pi}^{\pi} \cos(m+n)x dx = 0$, therefore $\int_{-\pi}^{\pi} \cos(mx)\cos(nx) dx = \begin{cases} \pi & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$	• Most students are able to apply the compound angle formula on $\cos(mx + nx)$, but unable to follow through as they only consider the value of $\int_{-\pi}^{\pi} \sin(mx)\sin(nx) dx$ for only one of the cases $m \neq n$ or $m = n$
(iii)	There are $2a$ repeated graphs on $[-\pi, \pi]$. 	• Most students didn't get this answer • Most students didn't seem to realize that there are $2a$ of repeating graphs in the interval. • Many wrote $\int_{-\pi}^{\pi} \sin(ax) dx = 2 \int_0^{\pi} \sin(ax) dx$, which suggests that they know that they need to consider only the

	$\int_{-\pi}^{\pi} \sin(ax) dx = 2a \int_0^{\frac{\pi}{a}} \sin(ax) dx$ $= 2a \int_0^{\frac{\pi}{a}} \sin(ax) dx$ $= 2a \left[-\frac{\cos(ax)}{a} \right]_0^{\frac{\pi}{a}}$ $= 2a \left(-\frac{\cos \pi}{a} + \frac{\cos 0}{a} \right)$ $= 2a \left(\frac{1}{a} + \frac{1}{a} \right)$ $= 2a \left(\frac{2}{a} \right)$ $= 4$	positive parts of the $\sin(ax)$ graph, but they considered the region and period wrong.
(iv)	$\pi \int_{-\pi}^{\pi} [\sin(ax) + 1]^2 dx$ $= \pi \int_{-\pi}^{\pi} [\sin(ax)]^2 + 2 \sin(ax) + 1 dx$ $= \pi \int_{-\pi}^{\pi} [\sin(ax)]^2 dx + 2\pi \int_{-\pi}^{\pi} \sin(ax) dx + \pi \int_{-\pi}^{\pi} 1 dx$ $= \pi(\pi) + 2\pi(4) + \pi[x]_{-\pi}^{\pi} \quad (\text{by letting } m = n = a)$ $= 3\pi^2 + 8\pi$	- A lot of students didn't realize that you can use part (i) to integrate $\int_{-\pi}^{\pi} [\sin(ax)]^2 dx$ - Common mistakes here include - Thinking that $\pi \int_{-\pi}^{\pi} [\sin(ax)]^2 dx + \pi(2\pi)$ is the area required, - Writing $\int_{-\pi}^{\pi} [\sin(ax) + 1]^2 dx = \int_{-\pi}^{\pi} [\sin(ax)]^2 dx + \int_{-\pi}^{\pi} 1^2 dx$

5	Suggested Solution	Marker's Comments
(i)	$\frac{P(X = k + 1)}{P(X = k)} = \frac{\binom{n}{k+1} p^{k+1} (1-p)^{n-k-1}}{\binom{n}{k} p^k (1-p)^{n-k}}$ $= \frac{\frac{n!}{(k+1)!(n-k-1)!} p^{k+1} (1-p)^{n-k-1}}{\frac{n!}{(k)!(n-k)!} p^k (1-p)^{n-k}}$ $= \frac{1}{(k+1)} p$ $= \frac{1}{(n-k)} (1-p)$ $= \frac{(n-k)p}{(k+1)(1-p)}$ or $= \frac{n! p^{k+1} (1-p)^{n-k-1}}{(k+1)!(n-k-1)!} \times \frac{(k)!(n-k)!}{n! p^k (1-p)^{n-k}}$ $= \frac{p}{(k+1)} \times \frac{(n-k)}{(1-p)}$ $= \frac{(n-k)p}{(k+1)(1-p)}$	This is a show question. To get the full marks, you must show every step clearly, especially the simplification. Please see the left column on examples on how to show the simplifications.
(ii)	Assuming k is the mode, then $P(X = k + 1) < P(X = k)$. $\frac{P(X = k + 1)}{P(X = k)} < 1$ $\frac{\frac{7}{9}(99-k)}{(k+1)\left(1-\frac{7}{9}\right)} < 1$	Many students missed out the statement 'assuming k is the mode'.

$77 - \frac{7}{9}k < \frac{2}{9}k + \frac{2}{9}$ $k > 76\frac{7}{9}$ i.e. $P(X = 77) > P(X = 78) > P(X = 79) > \dots$ This result also implies k cannot be less than $76\frac{7}{9}$ or otherwise $P(X = k+1) > P(X = k)$. Thus, $k = 77$	Note the instruction on ‘using the result in part (i)’ so you should not be using the GC to find the answer.
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6	Suggested Solution	Marker’s Comments
(a)	$4! \times \binom{5}{4} \times 4! = 2880$ $4! \times 2^4 = 384$	This part is well answered in general. Some students failed to distinguish between ‘slotting’ and ‘alternate’.
(b)	$(4-1)! \times 2^4 = 96$	This part is not well answered. Many students could not translate the idea of ‘each of the 4 couples can swap their positions’ correctly, $4 \times 2!$ was often seen instead of $(2!)^4$.
(c)	$\frac{\binom{8}{4}\binom{4}{4}}{2!} - 1 = 35 - 1 = 34$	This part is not well answered. Division by $2!$ was often missing in a solution for groups with identical size. Solving by listing the cases is possible but inefficient.

7	Suggested Solution	Marker’s Comments
(i)	Let A be the number of young employees, out of 15, who would like to enroll in either infocomm technology programme or financial programme. $1 - 0.29 = 0.71$ $A \sim B(15, 0.71)$ $P(A = 12) = 0.182098 \approx 0.182$	- Students are to identify that event A follows a binomial distribution. - For students who did not identify the distribution, most obtained the correct answer using ‘Required Prob = $\binom{15}{12}(0.71)^{12}(0.29)^3 = 0.182$.’
(ii)	Let H be the number of young employees, out of 15, who would like to enroll in human resource programme. $H \sim B(15, 0.29)$	‘not less than 5’ equates to ‘≥ 5’. Common Mistakes: - Students used $P(H > 5)$ instead of $P(H \geq 5)$. - Students wrote $P(H \geq 5) = 1 - P(H \leq 5)$.

	$P(H \geq 5) = 1 - P(H \leq 4)$ $= 1 - 0.45003054997$ $= 0.45003$ ≈ 0.450	
(iii)	Let M be the number of young employees, out of 4, who would like to enroll in financial management programme. $M \sim B(4, p)$ $P(M = 1 \text{ or } 2) = 0.3$ $P(M = 1) + P(M = 2) = 0.3$ $\binom{4}{1} p(1-p)^3 + \binom{4}{2} p^2(1-p)^2 = 0.3$ $4p(1-p)^3 + 6p^2(1-p)^2 = 0.3$ By GC, $p \approx 0.0861$ or $p \approx 0.7243$ (rejected as $0.7243 > 0.71$). So $p = 0.0861$ (to 3 s.f.)	- Most students are able to rewrite the expression found in MF26 to form the required equation $\binom{4}{1} p(1-p)^3 + \binom{4}{2} p^2(1-p)^2 = 0.3$. - However many are unable to use the GC to solve the quartic equation. Most obtained only 1 answer which is the incorrect one. - For students who obtained more than 1 answer, quite a few students did not realise that $p \approx 0.7243$ is rejected as $p \leq 0.71$ (from context).

8	Suggested Solution	Marker's Comments
(i)	$a = 30$. $\frac{3}{k} \times \frac{2}{k-1} \times \frac{1}{k-2} = 0.05$ (the table). By GC, $k = 6$	You need to know that the probability is formulated without replacement.
(ii)	Out of the k (or 6) envelopes, every envelope has an equal chance of being selected by Joe.	
(iii)	Let X be the amount of money that Joe receives after winning a game. By GC, $E(X) = 16.5$ and $\text{Var}(X) = 36.45$. Since $n \geq 20$ is large, by Central Limit Theorem,	Expected amount here signifies the need to use Central Limit Theorem since the distribution for the amount is not given.

$\bar{X} \sim N\left(16.5, \frac{36.45}{n}\right)$ approximately.

$$P(16.5 - 2 \leq \bar{X} \leq 16.5 + 2) > 0.95$$

$$P(14.5 \leq \bar{X} \leq 18.5) > 0.95$$

Method 1 (GC Table, *good for small integer n*)

n	$P(14.5 \leq \bar{X} \leq 18.5)$
34	0.94659316
35	0.94998281
36	0.95314596

Least $n = 36$

Method 2 (Standardisation, *tedious but always working*)

$$P\left(\frac{14.5 - 16.5}{\sqrt{\frac{36.45}{n}}} \leq \frac{\bar{X} - 16.5}{\sqrt{\frac{36.45}{n}}} \leq \frac{18.5 - 16.5}{\sqrt{\frac{36.45}{n}}}\right) > 0.95$$

$$P\left(\frac{-2}{\sqrt{\frac{36.45}{n}}} \leq Z \leq \frac{2}{\sqrt{\frac{36.45}{n}}}\right) > 0.95$$

$$P\left(Z \leq \frac{-2}{\sqrt{\frac{36.45}{n}}}\right) < 0.025$$

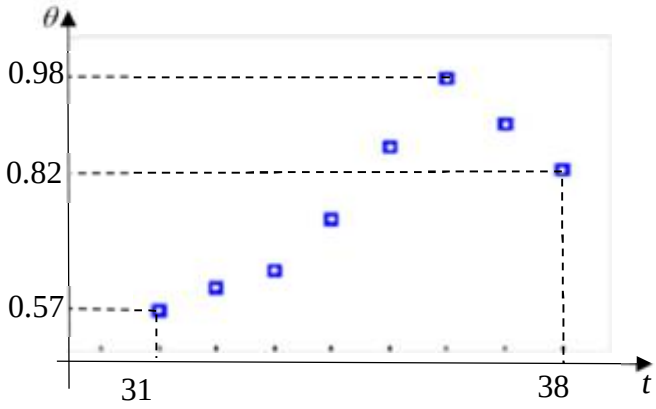
$$\frac{-2}{\sqrt{\frac{36.45}{n}}} < -1.96$$

$$\sqrt{\frac{36.45}{n}} > \frac{2}{1.96}$$

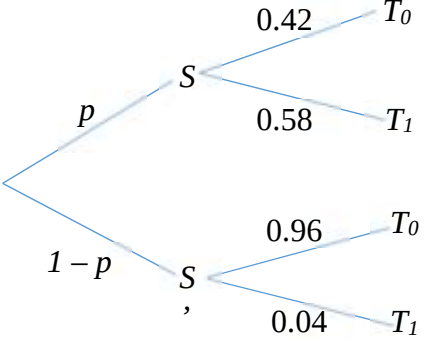
$$n > 35.00658$$

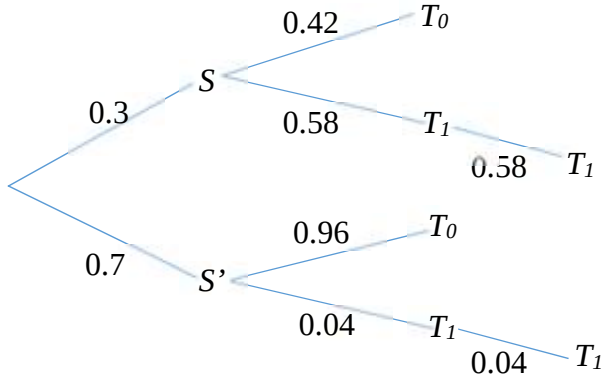
Least $n = 36$

Due to Central Limit Theorem, $\bar{X}$ follows a Normal Distribution APPROXIMATELY. The latter is an important keyword.

9	Suggested Solution	Marker's Comments
(i)		Well attempted except for some students did not indicate the range clearly on the scatter diagram. There are some who labelled the axes wrongly and did not use the variables provided in the context.
(ii)	$r = 0.851386 \approx 0.851$	
(iii)	The regression line θ on t should be used as t is the independent (or controlled) variable (Temperature varies with time)	Many students are able to choose the correct regression line with the correct reason.
(iv)	θ on t : $\theta = 0.0517857143t - 1.022857$ $\therefore \theta = 0.0518t - 1.02$ The temperature anomaly increases by approximately 0.0518°C every year.	Majority of students missed out the keyword “ approximately ”.
(v)	For Year 2025, $t = 2025 - 1980 = 45$, predicted value of $\theta = 0.0517857143(45) - 1.022857$ $= 1.307500$ ≈ 1.31 As $t = 45$ is not within the data range $[31, 38]$, the predicted value is not reliable.	Students did not realise that they need to state the data range of the given value.

(vi)	In many years to come, factors like various countries could have done more to reduce environment pollution, hence a linear model may not be suitable for long term predictions. OR In many years to come, if people are not putting in diligent effort to minimise environment pollution, the global temperature may change drastically hence a linear model may not be suitable for long term predictions. OR It is not possible for temperature anomaly to increase to an infinite value in long run.	Quite a number of students answered this part (taking into account of the context) well.
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10	Suggested Solution	Marker's Comments
(i)	Since conditional probabilities given the same event must sum up to 1, we have $q_1 = 0.96$ and $q_2 = 0.58$	Many students make an overly complicated attempt on this question when, they can actually just use the fact that $P(T_0 S) + P(T_1 S) = 1$ to deduce the values of q_1 and q_2. In this case, it means that the sum of the values in each column is 1.
(ii)	$ \begin{aligned} P(S T_0) &= \frac{P(S \cap T_0)}{P(T_0)} \\ &= \frac{0.42p}{0.42p + 0.96(1-p)} \\ &= \frac{0.42p}{0.42p + 0.96 - 0.96p} \\ &= \frac{0.42p}{0.96 - 0.54p} \\ &= \frac{0.06(7p)}{0.06(16 - 9p)} \\ &= \frac{7p}{16 - 9p} \end{aligned} $ 	Many students can evaluate $P(S \cap T_0)$ but didn't realize that $P(T_0) = P(T_0 S)P(S) + P(T_0 S')P(S')$.

(iii)	$\frac{d}{dp}(P(S T_0)) = \frac{d}{dp}\left(\frac{7p}{16-9p}\right)$ $= \frac{7(16-9p) - (-9)(7p)}{(16-9p)^2}$ $= \frac{112}{(16-9p)^2} > 0 \quad \text{for all } 0 < p < 1$ Therefore, $P(S T_0)$ is an increasing function of p. The higher the proportion of patients with heart disease, the higher is the probability that a patient has heart disease given a negative test result. OR If the majority of the population have heart disease, the negative test results are no longer indicative of the absence of heart disease in a patient.	Most students are able to apply quotient rule to get $\frac{112}{(16-9p)^2}$ and concluded that it must be positive. Common mistakes: - Some students think 2nd derivative is used to determine if a function is increasing when 1st derivative is the one they should check - Contextual interpretation of $P(S T_0)$ is not well done, the term p has to be explained. Furthermore, many students make a claim about $P(S T_0)$ increasing without the reference to increasing p.
(iv)	 <pre> graph LR Root(()) --- 0.3 S((S)) Root --- 0.7 Sp((S')) S --- 0.42 T0_1((T0)) S --- 0.58 T1_1((T1)) Sp --- 0.96 T0_2((T0)) Sp --- 0.04 T1_2((T1)) T1_1 --- 0.58 T1_1a((T1)) T1_2 --- 0.04 T1_2a((T1)) </pre>	Very badly done. While students realized that conditional probability is involved, the denominator is not correct. The event 'both test results are positive' are not broken into the events 'heart disease present and both test results are positive' and 'heart disease not present and both test results are positive' correctly. Same thing with the numerator, many students write $(0.3 \times 0.58)^2$ instead.

$ \begin{aligned} & P(\text{heart disease} \mid \text{both test results are positive}) \\ &= \frac{P(\text{heart disease} \cap \text{both test result are positive})}{P(\text{both test results are positive})} \\ &= \frac{P(\text{heart disease} \cap \text{both test results are positive})}{P(\text{heart disease} \cap \text{both test results are positive}) + P(\text{no heart disease} \cap \text{both test results are positive})} \\ &= \frac{0.3 \times 0.58 \times 0.58}{0.3 \times 0.58 \times 0.58 + 0.7 \times 0.04 \times 0.04} \\ &= \frac{0.10092}{0.10092 + 0.00112} \\ &= \frac{0.10092}{0.10204} \\ &\approx 0.98902 \\ &\approx 0.989 \end{aligned} $	
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11	Suggested Solution	Marker's Comments
(i)	Unbiased estimate of the population mean, $\bar{x} = \frac{38.9}{250} + 10 = 10.1556 \approx 10.2$ Unbiased estimate of the population variance, $s^2 = \frac{1}{249} \left(390.31 - \frac{38.9^2}{250} \right) \approx 1.543201 \approx 1.54$	Well-done for most students. Mistakes involved writing $\mu = \bar{x}$ and $\sigma^2 = s^2$.
(ii)	Test $H_0 : \mu = 10$ against $H_1 : \mu > 10$ Level of significance: 4% (upper-tailed) Under H_0, $\bar{X} \sim N\left(10, \frac{1.543201}{250}\right)$ approximately. $Z = \frac{\bar{X} - 10}{\sqrt{\frac{1.543201}{250}}} \sim N(0,1) \text{ approximately}$	Although students were generally able to obtain the correct p-value or z test statistic value from the GC, some had clearly shown that they understood little of the concepts underlying hypothesis testing by making the following mistakes: - Test $H_0 : x = 10$ against $H_1 : x > 10$ - Applying CLT to say that $\bar{X}$ follows a normal distribution because of the large sample size (this is redundant as X is already assumed to be normally distributed by the question).

	From GC, $p\text{-value} \approx 0.0238 < 0.04$ OR critical value, $z_{0.96} \approx 1.75$ and observed test statistic $\approx 1.98 > 1.75$ Thus, we reject H_0 and conclude that there is sufficient evidence at the 4 % level of significance that the mean speed of PMD riders exceeds the new speed limit.	- $\bar{X} \sim N\left(10.2, \frac{1.543201}{250}\right)$ and thus failed to understand that $E(\bar{X}) = \mu = 10$ (under H_0) instead of the sample mean 10.2. - Using the 3 s.f. values in part (i) to calculate the p-value or z test statistic that reduces the accuracy; students should bear in mind that at least 5 s.f. is needed for the use of intermediate values. - Writing p-value as p.
(iii)	There is a probability of 0.04 (or 4% chance) of wrongly rejecting the claim that the mean speed of PMD riders does not exceed (or is) 10 km/h.	Generally well-done.
(iv)	Test $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$ Level of significance: 4% (2-tailed) For a 2-tailed test, $p\text{-value} \approx 2(0.0238) = 0.0476 > 0.04$ The conclusion will change.	Badly done. Most students wrongly concluded that $p\text{-value} = 0.0238 > \frac{1}{2}(0.04) = 0.02$. The intention might have been to compare only the right tail but students should show that p-value is doubled in a 2-tailed test rather than ‘remaining as the same value’ as in part (ii).
(v)	Let n be the sample size. $s^2 = \frac{n}{n-1}(3.2^2) = \frac{10.24n}{n-1}$ Test $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$ Under H_0, $\bar{X} \sim N\left(10, \frac{10.24}{n-1}\right)$ approximately. Test statistic: $Z = \frac{\bar{X} - 10}{\sqrt{\frac{10.24}{n-1}}} \sim N(0,1)$ approximately	Badly done although a similar question had been done in the assignment. The main mistake was treating the given value 3.2 as population standard deviation instead of sample standard deviation, and hence wrongly writing $s^2 = \frac{3.2^2}{n}$. Surprisingly, majority of the students also wrongly wrote $\bar{X} \sim N\left(9.5, \frac{3.2^2}{n}\right)$ where they thought $E(\bar{X}) = 9.5$, although such an occurrence was less common in part (ii).

Level of significance: 5% (2-tailed) To reject H_0, $\frac{9.5-10}{\sqrt{\frac{10.24}{n-1}}} \leq -1.96 \quad \text{or} \quad \frac{9.5-10}{\sqrt{\frac{10.24}{n-1}}} \geq 1.96$ $\Rightarrow \sqrt{\frac{n-1}{10.24}} \geq 3.92 \quad \Rightarrow \sqrt{\frac{n-1}{10.24}} \leq -3.92 \text{ (N.A.)}$ $\Rightarrow n \geq 158.35$ $\Rightarrow n \geq 159$ Thus, smallest $n = 159$.	Many students only worked with the left tail although it's a 2-tailed test. This might result from observing that sample mean 9.5 is less than population mean 10, but an appropriate explanation for only considering the left tail should be stated in the solution in order not to compromise the evidence of understanding the rejection criteria of a 2-tailed test. Some fumbled with the inequality sign because $\bar{x} - \mu = 9.5 - 10 < 0$ and working with the right tail will result in no real solution. However, this group of students 'forced' their way through by writing statements like $\frac{-0.5}{\sqrt{\frac{10.24}{n-1}}} \geq 1.96 \Rightarrow \left(\frac{-0.5}{\sqrt{\frac{10.24}{n-1}}} \right)^2 \leq (1.96)^2$ to remove the negative sign, without realising that LHS is a negative value that cannot be greater than RHS which is a positive value and hence no solution. Thus, the idea of taking squares on both sides of an inequality is not well-understood.
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